

Learning and Decision-Making in Strategic Environments

Pre-Synopsis / Defense Seminar

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Introduction

Central Question

How should a learner act when agents respond strategically to its decisions, or when data is partitioned across non-communicating institutions?

Strategic Behavior

- Agents manipulate features, withhold information, or alter presentation to obtain preferred decisions.
- The deployed rule actively shapes what the learner observes.
- Requires game-theoretic modeling of the interaction between learner and agents.

Partitioned Observations

- Separate institutions observe (X, Y) and (X, Z) independently.
- The dependence between target Y and auxiliary variable Z is never observed jointly.
- Requires robust imputation and regularization to balance reliance on side information.

Two-Stage Framework: Decision & Learning

Across all problem settings, we employ a unified two-stage methodology:

Stage 1: The Decision Problem (Known Environment)

- Specify receiver, responding agent, and objectives under complete information.
- Formulate as a Stackelberg game: receiver commits to a rule, sender best responds.
- Characterize structural properties of optimal rules (penalty regions, probability ramps).

Stage 2: The Learning Problem (Unknown Environment)

- The decision maker observes only finite training samples or sequential online feedback.
- Exploit the game structure to construct data-driven algorithms and modified hypothesis classes.
- Establish theoretical sample complexity bounds, PAC learnability, or sublinear regret.

Strategic Classification

Setting: University Admissions with Strategic Applicants

- A university admits candidates to **Theoretical Physics (+)** and **Experimental Physics (−)** via a declared entrance exam.
- Every applicant has a **true aptitude** aligning with a **true program**.
- Applicants have personal **preferences** that may not match their true aptitude.
- Knowing the admission rule, students can **invest coaching effort** (incurring a cost) to manipulate their demonstrated score towards their preferred program.

Core Decision Question

"Given that applicants strategically game their scores, how can the university design an optimal decision rule that maximizes admissions to their true programs?"

Key Challenge: The university observes demonstrated scores, not true aptitudes. Standard naive cutoffs invite heavy manipulation and high misclassification.

Abstract Game Formulation (No specific sets or cost forms assumed)

Receiver (Leader)

- Feature space $\mathcal{X}$, label space $\mathcal{Y}$, true target $h : \mathcal{X} \rightarrow \mathcal{Y}$.
- Commits to rule $r : \mathcal{X} \rightarrow \mathcal{Y} \cup \{\Delta\}$, where Δ is a **penalty label** (or reject option).
- **Objective:** Minimize post-response error:

$$\min_r \mathbb{P}_{x \sim D}(r(s_r(x)) \neq h(x)).$$

Sender (Follower)

- True feature $x \in \mathcal{X}$ drawn from prior D .
- Observes r ; reports feature $x' \in \mathcal{X}$ maximizing net payoff:

$$s_r(x) \in \arg \max_{x' \in \mathcal{X}} [U(r(x'), x) - C(x, x')].$$

- $U(\Delta, x) = -\infty$; $C(x, x) = 0$.
- Ties broken in worst case against receiver.

Stackelberg Equilibrium Concept

A Stackelberg equilibrium rule r^* anticipates the follower's worst-case best response $s_r(x)$ and optimizes receiver classification accuracy over all commitments $r \in \mathcal{R}$.

Specifics of the Discrete Setting

- **Feature Space:** A finite set $\mathcal{X} = \{1, 2, \dots, n\}$.
- **True Function:** $h : \mathcal{X} \rightarrow \mathcal{Y} = \{-, +\}$ partitions points into two classes.
- **Sender Utility:** Normalized such that getting the true label yields zero utility: $U(h(x), x) = 0$, $U(\Delta, x) = -\infty$, and $U(y, x) \geq 0$ for $y \neq h(x)$.
- **Manipulation Cost:** General cost matrix $C : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ with $C(x, x) = 0$.
- **Receiver Strategy:** Deterministic assignment $r : \mathcal{X} \rightarrow \{-, +, \Delta\}$.

The Strategic Conflict on Discrete Domains

An agent of true type $\mathbf{i}$ ($h(\mathbf{i}) = -$) will impersonate report $\mathbf{l}$ intended for type $\mathbf{j}$ ($h(\mathbf{j}) = +$) whenever the utility gain exceeds the cost difference:

$$U(h(\mathbf{j}), \mathbf{i}) - C(\mathbf{l}, \mathbf{i}) + C(\mathbf{k}, \mathbf{i}) \geq 0.$$

Goal: Identify which points can be safely classified and which must receive penalty label Δ .

Definition 2.4 (Sender's Conflict Graph G_h)

Let $G_h = (V, E)$ with $V = \mathcal{X}^2$; node $(\mathbf{i}, \mathbf{k})$ denotes true feature $\mathbf{i}$ reporting $\mathbf{k}$. Nodes $(\mathbf{i}, \mathbf{k})$ and $(\mathbf{j}, \mathbf{l})$ have an undirected edge iff:

- (a) $\mathbf{i} = \mathbf{j}, \mathbf{k} \neq \mathbf{l}$ (unique reporting), or $h(\mathbf{i}) \neq h(\mathbf{j}), \mathbf{k} = \mathbf{l}$ (label clash);
- (b) $h(\mathbf{i}) \neq h(\mathbf{j})$ and manipulation incentive holds: $U(h(\mathbf{j}), \mathbf{i}) - C(\mathbf{l}, \mathbf{i}) + C(\mathbf{k}, \mathbf{i}) \geq 0$ (or vice versa).

Theorem 2.5 (Stackelberg Equilibrium $\iff$ Maximum Independent Set)

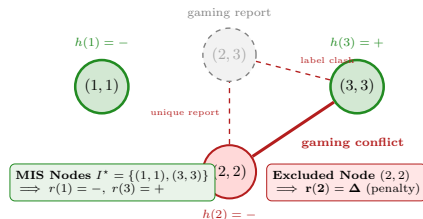
- (a) If I^* is a **maximum independent set** (MIS) of G_h , then $r^*(\mathbf{k}; I^*) = h(\mathbf{i})$ if $\exists \mathbf{i}$ s.t. $(\mathbf{i}, \mathbf{k}) \in I^*$, and Δ otherwise, is a Stackelberg equilibrium.
- (b) Conversely, any Stackelberg equilibrium r^* defines an MIS with $|I^*| = \alpha(G_h)$.

Combinatorial Takeaway

Optimal discrete classification reduces to finding an MIS on G_h ; points not in the MIS receive penalty Δ .

Illustrative 3-Point Setup & Conflict Graph:

- $\mathcal{X} = \{1, 2, 3\}$, with $h(1) = h(2) = -$ and $h(3) = +$.
- Cost $C(i, j) = |i - j|$; utility $U(+, 2) = 1.5$.
- Type 2 games: $U(+, 2) - C(3, 2) = 0.5 > 0$.
- Node $(2, 2)$ conflicts with $(3, 3)$ via gaming edge.
- MIS is $I^* = \{(1, 1), (3, 3)\}$ ($\alpha(G_h) = 2$).
- **Optimal Rule:** $r(1) = -, r(3) = +, r(2) = \Delta$.



Conflict graph G_h for 3-point domain (Thesis Sec. 2.3.1)

Contrast: Naive vs. Strategic Rule

Naive rule h (no Δ): Type 2 flips to 3 $\implies$ **Loss = 1.**

Strategic rule r^* : $r(2) = \Delta$ deters gaming $\implies$ **Loss = 0** on $\{1, 3\}$!

Specifics of the Continuous 1D Setting

- **Domain:** $\mathcal{X} = [0, 1]$, continuous distribution D with density $f_D(x) > 0$.
- **True Classifier:** Threshold classifier with class boundary $\mathbf{h} \in (0, 1)$:

$$h(x) = \begin{cases} - & x \in [0, \mathbf{h}], \\ + & x \in (\mathbf{h}, 1]. \end{cases}$$

- **Non-Uniform Preferences:** Senders have label-switching utilities: $U(+, x) = \mathbf{u}_+$ ($x \leq \mathbf{h}$), $U(-, x) = \mathbf{u}_-$ ($x > \mathbf{h}$), $U(\Delta, x) = -\infty$.
- **Manipulation Cost:** Cost kernel $C(x, x') = |\phi(x') - \phi(x)|$, ϕ strictly increasing.

Regimes of Strategic Behavior:

- **Cases 1–3 (Trivial):** Aligned preferences ($\mathbf{u}_+, \mathbf{u}_- \leq 0$) or manipulation too cheap ($\mathbf{u}_+ \geq 1$) $\implies$ truthful or trivial majority rule.
- **Cases 4 & 5 (Strategic Regimes):** $0 < \mathbf{u}_+ \leq 1$ and $\mathbf{u}_- \leq \mathbf{u}_+$. Manipulation is actively controlled by deploying the **penalty label Δ** .

Theorem 2.10 (Deterministic Stackelberg Equilibrium — Case 4)

Let $0 < \mathbf{u}_+ \leq 1$ and $-\mathbf{u}_+ < \mathbf{u}_- \leq \mathbf{u}_+$. Define cutoffs $\mathbf{h}^-, \mathbf{h}^+$ such that $0 \leq \mathbf{h}^- \leq \mathbf{h} < \mathbf{h}^+ \leq 1$ and $\phi(\mathbf{h}^+) - \phi(\mathbf{h}^-) = \mathbf{u}_+$. Then the receiver strategy

$$r(x; \mathbf{h}^-, \mathbf{h}^+) = \begin{cases} h(x) & x \in [0, \mathbf{h}^-] \cup [\mathbf{h}^+, 1], \\ \Delta & x \in (\mathbf{h}^-, \mathbf{h}^+), \end{cases}$$

is a Stackelberg equilibrium. The optimal pair $(\mathbf{h}^-, \mathbf{h}^+)$ minimizes the mass $D((\mathbf{h}^-, \mathbf{h}^+))$.

Theorem 2.11 (Case 5: Zero Classification Error!)

When sender preferences are strongly opposing ($1 \geq \mathbf{u}_+ > 0$ and $\mathbf{u}_- \leq -\mathbf{u}_+$), setting $\mathbf{h}^- = \mathbf{h}$ and $\phi(\mathbf{h}^+) - \phi(\mathbf{h}) = \mathbf{u}_+$ achieves **Zero Classification Error**: $\mathcal{E}(r^*) = 0$.

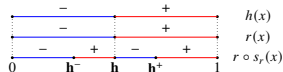
Mechanism: Receiver creates a *buffer zone* of width $\mathbf{u}_+$ around the boundary. The penalty region deters crossing and strictly confines misclassification.

Continuous Setting: Example & Naive vs. Strategic Gain

Canonical Setting: Uniform D on $[0, 1]$, $\phi(x) = x$, boundary $\mathbf{h} = 0.5$, symmetric utility $2\mathbf{u}$.

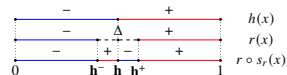
1. Naive Rule $h(x)$ (Passive)

- Boundary placed at $\mathbf{h} = 0.5$.
- Senders in $(\mathbf{h} - 2\mathbf{u}, \mathbf{h}]$ flip to $+$; senders in $(\mathbf{h}, \mathbf{h} + 2\mathbf{u}]$ flip to $-$.
- Misclassification over $(\mathbf{h} - 2\mathbf{u}, \mathbf{h} + 2\mathbf{u})$:
 $\mathcal{E}(\mathbf{h}) = 4\mathbf{u}$.



2. Strategic Rule r^* (With Penalty Δ)

- Sets buffer $(\mathbf{h} - \mathbf{u}, \mathbf{h} + \mathbf{u})$ to Δ .
- Penalty $-\infty$ prevents cross-boundary gaming.
- Misclassification strictly $[\mathbf{h} - \mathbf{u}, \mathbf{h} + \mathbf{u}]$:
 $\mathcal{E}(r^*) = 2\mathbf{u}$ (50% Reduction!)



University Admission Interpretation

Excluding boundary questions (difficulty in $[0.48, 0.52]$) prevents borderline candidates from gaming programs, cutting misclassification error strictly in half.

Higher Dimensions: Linear Boundary in $\mathbb{R}^n$ (Thesis Sec. 2.3.3)

- Feature space $\mathcal{X} \subset \mathbb{R}^n$ (bounded convex set, uniform D). Cost: $\|x - x'\|_2$.
- Ground truth: Hyperplane boundary $h(x) = +$ if $w^\top x > \mathbf{h}$, else $-$ (unit normal $w \in \mathbb{R}^n$).
- Boundary surface set $\mathcal{Z} = \{z \in \mathcal{X} : w^\top z = \mathbf{h}\}$ with normalized area $\sigma(\mathcal{Z}) = \frac{\mathcal{H}^{n-1}(\mathcal{Z})}{\text{Vol}(\mathcal{X})}$.

Theorems 2.12 & 2.13 (The Optimal Δ -Slab Equilibrium)

- **Lower Bound:** For any deterministic receiver rule r , $\mathcal{E}(r) \geq 2\mathbf{u} \cdot \sigma(\mathcal{Z})$.
- **Optimal Δ -Slab Strategy:** Parallel slab of width $2\mathbf{u}$ with penalty label Δ :
$$r^*(x) = + \text{ if } w^\top x \geq \mathbf{h} + \mathbf{u}; \quad - \text{ if } w^\top x \leq \mathbf{h} - \mathbf{u}; \quad \Delta \text{ if } |w^\top x - \mathbf{h}| < \mathbf{u},$$

achieving the lower bound with **exact equality**: $\mathcal{E}(r^*) = 2\mathbf{u} \cdot \sigma(\mathcal{Z})$.

Equivalence via Dimension Reduction

The scalar projection mapping $\mathcal{M}(x) = \frac{w^\top x - a_{\min}}{a_{\max} - a_{\min}}$ projects $\mathbb{R}^n \rightarrow [0, 1]$, proving that 1D structural properties and buffer geometry generalize directly to multi-dimensional linear classifications!

Specifics of the Stochastic Formulation

- Receiver strategy is randomized: $r(x) = (p(x), q(x), \delta(x)) \in \Delta(\{-, +, \Delta\})$.
- Sender chooses report x' to maximize expected payoff:

$$s_r(x) \in \arg \max_{x' \in [0,1]} \left[\mathbb{E}_{y \sim r(x')} [U(y, x)] - |\phi(x') - \phi(x)| \right].$$

Theorem 2.14 (Stochastic Stackelberg Equilibrium)

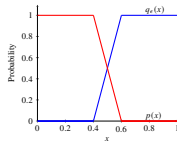
Let D be uniform on $[0, 1]$, $\phi(x) = x$, $\mathbf{u}_+ = \mathbf{u}_- = 2\mathbf{u}$, and define $\mathbf{h}^- = \mathbf{h} - \mathbf{u}$, $\mathbf{h}^+ = \mathbf{h} + \mathbf{u}$. The randomized decision rule with **no penalty label** ($\delta(x) \equiv 0$):

$$r^*(x) = \begin{cases} (1, 0, 0) & x \leq \mathbf{h}^-, \\ \left(1 - \frac{x - \mathbf{h}^-}{2\mathbf{u}}, \frac{x - \mathbf{h}^-}{2\mathbf{u}}, 0\right) & x \in (\mathbf{h}^-, \mathbf{h}^+), \\ (0, 1, 0) & x \geq \mathbf{h}^+, \end{cases}$$

is a Stackelberg equilibrium achieving expected error $\mathcal{E}(r^*) = \frac{\mathbf{u}}{2}$.

Mechanism: Why Randomization Works

- Probability $q(x)$ ramps linearly with slope $\frac{1}{2u}$.
- Marginal utility gain: $2u \times \frac{1}{2u} = 1$.
- Matches marginal cost of effort $\frac{\partial}{\partial x'} |x' - x| = 1$!
- Marginal gain matches cost everywhere $\implies$ truth-telling $s_r(x) = x$ is optimal!



Linear ramp $q(x)$ and $p(x)$ (Thesis Fig. 2.3)

	Naive Rule	Deterministic Δ -Rule	Stochastic Ramp Rule
Receiver Error	$4u$	$2u$	$u/2$
Relative Gain	Baseline (0%)	50% reduction	87.5% reduction!
Mechanism	Passive	Buffer region with Δ	Continuous probability ramp

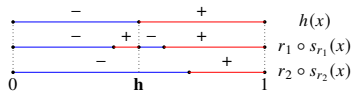
Insight: Randomization provides continuous leverage. The penalty label Δ compensates for the absence of randomization when deterministic rules are legally or practically required.

Equilibrium Multiplicity in Deterministic Rules:

Any interval $[h^-, h^+]$ of length $2u$ containing $h = 0.5$ achieves minimal error $2u$, but structures differ:

Rule 1 (Symmetric Buffer $[0.45, 0.55]$):

- Equal errors on both classes $\implies$ **Fair**.
- Induced classifier $r_1 \circ s_{r_1}$ has **3 boundaries**
 $\implies$ higher VC dimension!



Thesis Fig. 2.4: Induced boundaries under r_1 vs r_2

Rule 2 (Asymmetric Buffer $[0.40, 0.50]$):

- Errors only on class $- \implies$ **Unfair**.
- Induced classifier $r_2 \circ s_{r_2}$ has **2 boundaries**
 $\implies$ lower VC dimension!

Fundamental Trade-off

Enforcing **group fairness** increases the boundary complexity (VC dimension), requiring higher sample complexity to learn!

The Principal-Agent Perspective: Preventing Degenerate Rules

- Minimizing error without constraints could lead to degenerate rules (e.g., $r(x) = \Delta$ a.e.).
- Senders in practice have an outside **reservation payoff** ν .

Negative Reservation ($\nu \leq 0$)

- Senders opt out only if suffering negative payoff.
- Degenerate rules yield $-\infty \implies$ sender rejects.
- Symmetric buffer r_1 satisfies participation: $\min_x \mathbb{U}(x; r_1, s_{r_1}) = 0 \geq \nu$.

Positive Reservation ($\nu > 0$)

- Senders require a positive baseline reward.
- Truthful reporting yields utility $0 < \nu$, so participation constraints bind tightly across domain.
- Rules must balance deterrence with participation.

Takeaway

Participation constraints rule out degenerate penalty assignments and provide an economic foundation for selecting non-degenerate buffer equilibria.

The Learning Paradigm: Unknown Environment (Thesis Sec. 2.4)

- Receiver knows game structure (C, U) , but not ground truth h or distribution D .
- Observes finite training sample $S = \{(x_i, y_i)\}_{i=1}^m \stackrel{\text{i.i.d.}}{\sim} D$.
- **Strategic PAC Learnability (Def. 2.17):** Outputs $\hat{r}$ such that with prob $> 1 - \delta$:

$$\mathcal{E}(\hat{r}; h, D) - \mathcal{E}(r_{h,D}^*; h, D) < \epsilon \quad \text{for } m \geq m_{\mathcal{F}}(\epsilon, \delta).$$

Multiple Decision Boundaries: Generalizing to $VC(\mathcal{F}) = d$

- **Hypothesis Class:** $\mathcal{F} \subset \{f : [0, 1] \rightarrow \{-, +\}\}$ with at most $d - 1$ decision boundaries ($\implies VC(\mathcal{F}) = d$).
- **Stackelberg Structure (Prop. 2.16):** For intervals $O_1, \dots, O_{d-1}$, optimal rule assigns active sub-intervals $\Theta_n \subseteq O_n$ to $h(x)$ and intervening buffer zones to penalty Δ .
- Guarantees that multi-boundary 1D problems preserve piecewise Stackelberg structure.

Algorithm 1: Vanilla ERM (Two-Stage)

- 1: **Input:** $\mathcal{F}$, sample $S = \{(x_i, y_i)\}_{i=1}^m$
- 2: **Estimation:** Fit non-strategic rule:

$$\hat{h} \leftarrow \arg \min_{f \in \mathcal{F}} \sum_{i=1}^m \mathbb{1}_{\{f(x_i) \neq y_i\}}$$

- 3: Estimate empirical law $D(S)$
- 4: **Decision:** Deploy Stackelberg rule:

$$\hat{r} \leftarrow r_{\hat{h}, D(S)}^* \quad (\text{plug-in})$$

- 5: **Output:** $\hat{r}$

Philosophy: Decouples estimation from game; ignores gaming during training.

Algorithm 2: Strategy-Aware ERM (SERM)

- 1: **Input:** $\mathcal{F}$, sample $S = \{(x_i, y_i)\}_{i=1}^m$
- 2: **Transform:** Stackelberg class:

$$\tilde{\mathcal{F}} \leftarrow \{r_{f,D}^* \circ s_{r_{f,D}^*} : f \in \mathcal{F}, D \in \mathcal{D}\}$$

- 3: **Strategic ERM:** Optimize directly:

$$\hat{r} \circ s_{\hat{r}} \leftarrow \arg \min_{\tilde{f} \in \tilde{\mathcal{F}}} \sum_{i=1}^m \mathbb{1}_{\{\tilde{f}(x_i) \neq y_i\}}$$

- 4: **Output:** $\hat{r}$

Philosophy: Embeds game incentives directly into empirical risk optimization.

Fundamental Theorems of Strategic Learning

- **Theorem 2.21 (Equivalence):** Let $\mathcal{F}$ have $\leq d - 1$ boundaries. Then $\mathcal{F}$ is **strategically learnable under Strategy-Aware ERM** if and only if

$$\text{VC dim}(\tilde{\mathcal{F}}) < \infty.$$

- **Corollary 2.22 (Hierarchy):** For standard $\mathcal{F}$ and induced Stackelberg class $\tilde{\mathcal{F}}$:

$$\text{VC dim}(\mathcal{F}) < \infty \implies \text{VC dim}(\tilde{\mathcal{F}}) < \infty.$$

Implication: Any class learnable by Vanilla ERM is also learnable by Strategy-Aware ERM.

Sample Complexity Comparison

- **Vanilla ERM:** Sample complexity $m(\epsilon, \delta) = \Theta\left(\frac{d \log(1/\epsilon) + \log(1/\delta)}{\epsilon}\right)$ via DKW theorem.
- **Strategy-Aware ERM:** Sample complexity governed directly by $\tilde{d} = \text{VC dim}(\tilde{\mathcal{F}})$.

The Asymmetry Punchline: $VC(\mathcal{F}) = \infty \not\Rightarrow VC(\tilde{\mathcal{F}}) = \infty$ (Example 2.23)

- Consider rational-subset hypothesis class $f_F(x)$ where $F \subseteq ([f, 1] \cap \mathbb{Q})$.
- Class can shatter arbitrarily large subsets of rationals $\Rightarrow VC(\mathcal{F}) = \infty$! Vanilla ERM fails.
- Yet strategic penalty deterrence collapses equilibrium to constant rule $r \equiv - \Rightarrow VC(\tilde{\mathcal{F}}) = 0$!
- **Punchline:** Strategic incentives can compress infinite complexity into finite learnability!

Finite-Dimensional Regime Selection (Example 2.24):

Wide Intervals ($k > 1$)

- Class boundaries expand to $3n$.
- $VC(\tilde{\mathcal{F}}) \geq VC(\mathcal{F})$.
- **Recommendation: Vanilla ERM** is simpler and more sample-efficient.

Narrow Intervals ($k \leq 1$)

- Minority intervals collapse to Δ .
- Boundaries drop to $1 \ll n$.
- **Recommendation: SERM** dramatically compresses learning!

Learning with Side Information

Setting: Available Test-Time Side Information from Partitioned Data

- At test time, an auxiliary variable $Z \in \{-1, +1\}$ is available alongside features X to predict Y .
- **Example:** University admissions (Y) aided by a secondary college decision (Z), or primary hospital diagnosis assisted by secondary laboratory testing.

Partitioned Training Data:

$$D = \{(x_i, y_i)\}_{i=1}^n, \quad S = \{(x_j, z_j)\}_{j=1}^m.$$

- Disjoint institutional silos.
- Target Y and side info Z are **never observed jointly!**

Test-Time Deployment:

Predict outcome Y using joint (X, Z) .

- Full information (X, Z) is available.
- Learner must bridge data partition.

Core Decision Question

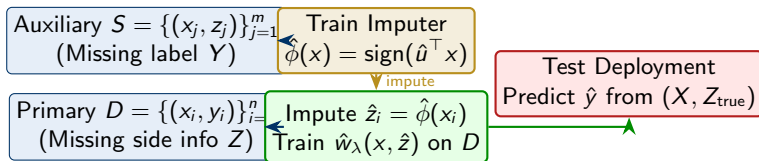
“How can a learner optimally exploit test-time side information Z when its joint relationship with Y was never observed during training?”

Partitioned Data: Formulation & Imputation Process

$(X, Y, Z) \sim \mathcal{D}$ on $\mathcal{X} \times \{-1, 1\}^2$. Regularity conditions (C1)–(C4) hold (boundedness, positive definiteness, non-separability).

Classifiers (Logistic Loss $\ell(a, b; c) = -\ln(\sigma(b \cdot c^\top(1, a)))$):

- **Secondary Imputer:** $\hat{u} \leftarrow \arg \min_u \hat{L}(u) + \mu \|u\|^2$ on $S \implies \hat{\phi}(x) = \text{sign}(\hat{u}_c + \hat{u}_x^\top x)$ maps $X \rightarrow Z$.
- **Target Composite:** $\hat{w}_\lambda \leftarrow \arg \min_w \hat{R}^{\hat{\phi}}(w) + \lambda_c |w_c|^2 + \lambda_x \|w_x\|^2 + \lambda_z |w_z|^2$ on imputed D .



Algorithm 3.1: Partitioned Side Information Learning (PSIL)

Require: Primary dataset $D = \{(x_i, y_i)\}_{i=1}^n$, auxiliary dataset $S = \{(x_j, z_j)\}_{j=1}^m$

Ensure: Composite classifier $\hat{w}_\lambda : \mathcal{X} \times \mathcal{Z} \rightarrow \mathcal{Y}$

- 1: **Step 1 (Train Imputer):** Learn $\hat{u} = \arg \min_u \hat{L}(u) + \frac{c}{\sqrt{m}} \|u\|^2$ on S ; set $\hat{\phi}(x) = \text{sign}(\hat{u}_c + \hat{u}_x^\top x)$.
- 2: **Step 2 (Data Imputation):** For each $(x_i, y_i) \in D$, compute imputed side information $\hat{z}_i \leftarrow \hat{\phi}(x_i)$.
- 3: **Step 3 (Regularized Learning):** Set $\lambda_c = \lambda_x = 1/\sqrt{n}$, and choose $\lambda_z \geq 1/\sqrt{n}$. Compute

$$\hat{w}_\lambda \leftarrow \arg \min_{w \in \mathbb{R}^{d+2}} \hat{R}^{\hat{\phi}}(w) + \lambda_c |w_c|^2 + \lambda_x \|w_x\|^2 + \lambda_z |w_z|^2.$$

- 4: **Step 4 (Deployment):** Given test point (x, z) with observed z , predict $\hat{y} = \text{sign}(\hat{w}_{\lambda,c} + \hat{w}_{\lambda,x}^\top x + \hat{w}_{\lambda,z} z)$.

Crucial Role of λ_z : Controls trust in imputed side information. Tuning λ_z balances exploiting Z 's predictive signal against protecting against imputation error in $\hat{\phi}$.

Theorem 3.6 (Excess Risk Upper Bound)

Under (C1)–(C4), (ND), let $\lambda_c = \lambda_x = 1/\sqrt{n}$ and $\lambda_z \geq 1/\sqrt{n}$. With probability at least $1 - \delta$:

$$R(\hat{w}_\lambda) - R(w^*) \leq \frac{C_1}{\sqrt{n}} + \min\{\lambda_z |w_z^*|^2, \Delta\} + \min\left\{\frac{1}{2\lambda_z}, W\right\} \frac{4}{\ln 2} \left(\frac{C_2}{\sqrt{m}} + L(u^*)\right),$$

where $\Delta = T(v^*) - R(w^*) \geq 0$ is the value of side info, and W is the population weight bound.

Four-Term Error Decomposition:

$$R(\hat{w}_\lambda) - R(w^*) = \underbrace{R(\hat{w}_\lambda) - R^{\hat{\phi}}(\hat{w}_\lambda)}_{\text{(A): Imputation gap at } \hat{w}_\lambda} + \underbrace{R^{\hat{\phi}}(\hat{w}_\lambda) - R^{\hat{\phi}}(w_{\hat{\phi},\lambda}^*)}_{\text{(B): Empirical estimation}} + \underbrace{R^{\hat{\phi}}(w_{\hat{\phi},\lambda}^*) - R(w_{\hat{\phi},\lambda}^*)}_{\text{(C): Imputation gap at } w_{\hat{\phi},\lambda}^*} + \underbrace{R(w_{\hat{\phi},\lambda}^*) - R(w^*)}_{\text{(D): Regularization penalty}}$$

- **(A) + (C) [Imputation Error]:** $\leq \min\{\frac{1}{2\lambda_z}, W\} \frac{4}{\ln 2} (\frac{C_2}{\sqrt{m}} + L(u^*)) \implies$ controlled by auxiliary sample m and protected by λ_z .
- **(B) + (D) [Estimation & Regularization]:** $\leq \frac{C_1}{\sqrt{n}} + \min\{\lambda_z |w_z^*|^2, \Delta\} \implies$ controlled by primary sample n .

Corollary 3.8 (The Three-Candidate Regularization Rule)

The optimal regularizer λ_z^* minimizing the upper bound belongs to **three candidates**:

$$\lambda_0 = \frac{1}{\sqrt{n}} \text{ (trust imputed } Z), \quad \lambda_\infty = \infty \text{ (ignore } Z, \text{ X-only model)}, \quad \lambda_I = \sqrt{\frac{b_m}{2a}} \text{ (interior balance)}.$$

Governed by **signal-to-imputation ratio** $\rho = |w_z^*|^2 / L(u^*)$: selects λ_∞ if $\rho < \rho_{\text{low}}$ (noisy Z), λ_0 if $\rho > \rho_{\text{high}}$ (informative Z), and λ_I otherwise.

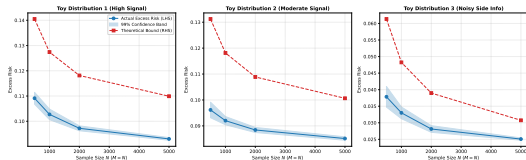
Theorem 3.7 (Minimax Lower Bound: Irreducible Observation Cost)

Let (D, S) be partitioned data and $A(D, S)$ be **any** learning algorithm. Then $\exists \mathcal{D}$ satisfying (C1)–(C4) such that:

$$\mathbb{P}\left(\mathbf{R}(\mathbf{A}(\mathbf{D}, \mathbf{S})) - \mathbf{R}(\mathbf{w}^*) > 0.19\right) \geq \frac{1}{2}.$$

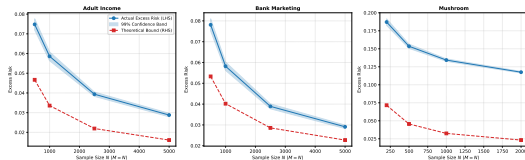
Implication: Even with infinite samples ($n, m \rightarrow \infty$), excess risk remains strictly positive due to an **information-theoretic barrier** that no algorithm can overcome.

Synthetic GMM Toy Distributions



Excess risk across 3 signal regimes ($m = n$, 20 runs, 99% CI). Actual excess risk closely tracks theoretical bound decay.

Real-World Benchmark Datasets



OpenML datasets (Adult Income, Bank Marketing, Mushroom). Demonstrates consistent $1/\sqrt{n}$ decay trajectory across diverse real domains.

Strategic Feature Revelation

Context: Strategic Information Disclosure

- In many high-stakes interactions, agents strategically withhold or disclose verifiable evidence (e.g., job candidates presenting selective test scores, loan applicants disclosing assets, firms reporting ESG metrics).
- **Verifiability:** Features cannot be fabricated, but unfavorable features can be strategically concealed.

The Revelation Regimes

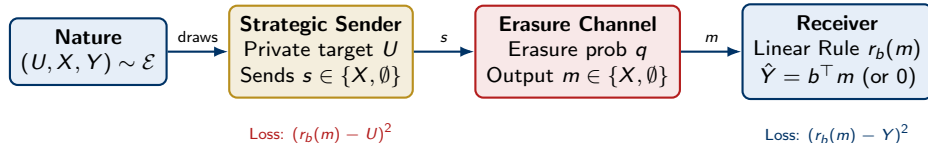
- **Mandatory Disclosure (Vanilla Baseline):** Receiver compels disclosure of all features; communication suffers from exogenous transmission loss / packet erasure with probability $q \in [0, 1)$.
- **Voluntary Disclosure (Strategic Regime):** Sender strategically decides whether to reveal X or withhold it, balancing private utility against receiver prediction.

The Central Question

“Can a sender’s strategic withholding of verifiable information ever be strictly advantageous to the receiver compared to mandatory disclosure?”

Strategic Feature Revelation: Problem Formulation

- **Information Structure:** Nature draws state $(U, X, Y) \sim \mathcal{E}$ where $X \in \mathcal{X} \subset \mathbb{R}^d$ (verifiable features), $Y \in \mathbb{R}$ (true target), and $U \in \mathbb{R}$ (sender's private preference target).
- **Receiver Commitment:** Commits to linear rule $r_b(x) = b^\top x$ on revealed features, and default $r_b(\emptyset) = 0$ when withheld or erased.
- **Sender Action:** Chooses message $s \in \{X, \emptyset\}$ to minimize expected squared error $(r_b(m) - U)^2$.
- **Erasure Channel:** Revealed features are erased with probability $q \in [0, 1)$:
 $\mathbb{P}(m = \emptyset \mid s = X) = q$.



Decision Problem: Sender Response & Strategic Advantage

Sender Best Response ($q < 1$):

- Sender reveals $s = X$ if $\mathbb{E}[(r_b(m) - U)^2 \mid s = X] < \mathbb{E}[(r_b(m) - U)^2 \mid s = \emptyset]$.
- $(1 - q)(b^\top X - U)^2 + qU^2 < U^2 \iff (2U - b^\top X)b^\top X > 0$.
- Defining $F_b(u, x) := (2u - b^\top x)b^\top x$, sender reveals iff $\mathbf{F}_b(\mathbf{U}, \mathbf{X}) > \mathbf{0}$.

Receiver Risks & Strategic Advantage:

- **Vanilla Risk:** $R_V(b) = q\mathbb{E}[Y^2] + (1 - q)\mathbb{E}[(Y - b^\top X)^2]$; optimal $b_V = \text{Var}(X)^{-1} \text{Cov}(X, Y)$.
- **Strategic Risk:** $R_S(b) = \mathbb{E}[Y^2] - (1 - q)\mathbb{E}[G_b(X, Y)\mathbb{I}\{F_b(U, X) > 0\}]$, with $G_b(x, y) := (2y - b^\top x)b^\top x$.

Strategic Advantage & Channel Invariance

$$\Delta(\mathcal{E}) := \inf_b R_V(b) - \inf_b R_S(b) = (1 - q)\Delta(0).$$

Key Invariance: The erasure channel scales the magnitude of Δ by $(1 - q)$, but **strictly preserves its sign** for all $q < 1$!

Conditions for Positive Strategic Advantage ($\Delta > 0$)

Assume centered variables and finite 2nd moments.

Define:

$$A(b) := \mathbb{E}|G_b| - \text{Var}(b_V^\top X) - \text{Var}((b - b_V)^\top X).$$

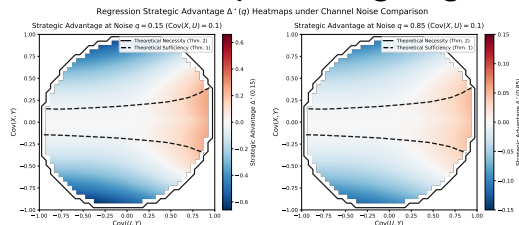
Theorems 4.3 & 4.4 (Advantage Conditions)

- **Necessary:** $\Delta(\mathcal{E}) > 0 \implies \exists b : A(b) > 0$.
- **Sufficient (Preference Alignment):**

$$\exists b : \|U - Y\|_2 < \frac{A(b)}{4\sqrt{\text{Var}(b^\top X)}} \implies \Delta(\mathcal{E}) > 0.$$

Interpretation: When preferences are aligned, withholding acts as an informative non-linear filter, improving prediction!

Parameter Sweep: Advantage Region



Parameter sweep across alignment $\|U - Y\|_2$ and variance, identifying the strictly positive strategic advantage regime ($\Delta(\mathcal{E}) > 0$).

Unknown Law, Finite Training Sample:

$D_n = \{(U_i, X_i, Y_i)\}_{i=1}^n \stackrel{\text{i.i.d.}}{\sim} \mathcal{E}$ with $\|X\|_2, |U|, |Y| \leq 1$.

SAR-ERM Estimators:

$$\hat{R}_V(b) = \frac{1}{n} \sum_{i=1}^n [qY_i^2 + (1-q)(Y_i - b^\top X_i)^2],$$

$$\hat{R}_S(b) = \frac{1}{n} \sum_{i=1}^n [Y_i^2 - (1-q)G_b(X_i, Y_i)\mathbb{I}\{F_b > 0\}].$$

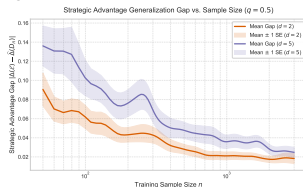
$$\hat{\Delta} = \inf_{\|b\|_2 \leq 1} \hat{R}_V(b) - \inf_{\|b\|_2 \leq 1} \hat{R}_S(b).$$

Theorem 4.6 (Uniform Generalization Bound)

With probability at least $1 - \delta$:

$$|\Delta(\mathcal{E}) - \hat{\Delta}| \leq \tilde{\mathcal{O}} \left(\frac{d^3}{\sqrt{n}} + \frac{d^2 \sqrt{\log(1/\delta)}}{\sqrt{n}} \right).$$

Empirical Generalization Gaps



Error gaps $|\Delta - \hat{\Delta}|$ across sample sizes n , confirming parametric $\mathcal{O}(1/\sqrt{n})$ rate.

Sequential Interaction Protocol: At each round $t = 1, \dots, T$:

- ① Receiver commits to joint action $a_t = (m_t, b_t) \in \{V, S\} \times \mathcal{B}_d(1)$ (regime and slope).
- ② Nature draws $(U_t, X_t, Y_t) \sim \mathcal{E}$; sender reveals if $m_t = V$ or $(m_t = S \text{ and } F_{b_t} > 0)$.
- ③ Receiver observes **bandit loss feedback**: single realized squared error $Z_t = \ell(r_{a_t}, Y_t) + \eta_t$.

Composite Kernel Gaussian Process Thompson Sampling (GP-TS):

- Model expected loss $f(m, b) = \mathbb{E}[Z \mid a = (m, b)]$ with composite Matérn-5/2 covariance:

$$k((m, b), (m', b')) = \mathbb{1}_{\{m=m'\}} \cdot k_{5/2}(b, b').$$

Decouples vanilla and strategic regimes while preserving shared smoothness properties.

- **Algorithm:** Maintain posterior $\mathcal{GP}(\mu_{t-1}, k_{t-1})$; draw inflated sample $\tilde{f}_t \sim \mathcal{GP}(\mu_{t-1}, v_t^2 k_{t-1})$; select $a_t = \arg \min_{a \in \mathcal{A}_t} \tilde{f}_t(a)$ over discretization net; observe Z_t and update posterior.

Theorem 4.10 (Sublinear Cumulative Regret)

Assume regime risks lie in Matérn-5/2 RKHS with $\|f\|_{\mathcal{H}} \leq B$, and noise is conditionally sub-Gaussian. Then GP-TS achieves:

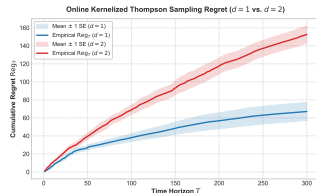
$$\text{Reg}_T = \sum_{t=1}^T [f(a_t) - \inf_a f(a)] = \tilde{O}\left(T^{\frac{5+3d}{10+2d}}\right).$$

Sublinear Exponent Across Dimension:

$d = 1$	$d = 2$	$d = 3$	$d = 4$	$d \geq 5$
$T^{0.67}$	$T^{0.79}$	$T^{0.88}$	$T^{0.94}$	Linear

Takeaway: Sublinear regret $\text{Reg}_T / T \rightarrow 0$ holds for all $d \leq 4$, establishing online learnability.

Cumulative Regret Trajectories



Cumulative regret curves for $d = 1, 2$ across rounds T , exhibiting sublinear empirical scaling in accordance with Theorem 4.10.

Strategic IPO Regulation

Economic Context: Initial Public Offerings (IPOs)

- Private firms raise public equity capital by selling shares to retail and institutional investors.
- **Information Asymmetry:** Issuing firms possess private knowledge of true productivity $Y(X)$, but present self-selected marketing disclosures, metrics, and projections z .

Regulatory Dilemma: The Tolerance Parameter ε

- Regulators enforce disclosure standards restricting presentation deviation: $\|z - X\|_2 \leq \varepsilon$.
- **Zero Tolerance** ($\varepsilon \rightarrow 0$): Eliminates misrepresentation, but creates severe compliance frictions and suppresses capital formation.
- **Lax Tolerance** ($\varepsilon \rightarrow \infty$): Maximizes trading liquidity, but induces severe lemon-market adverse selection and misallocation.

The Central Policy Question

“How should a regulator set tolerance ε to balance capital formation against adverse selection when firms strategically window-dress their presentations?”

Game Primitives:

- **Firm:** Fundamentals $X \in \mathcal{X} \subset \mathbb{R}^d$, true productivity $Y(X) \in \mathbb{R}_+$.
- **Regulator:** Announces tolerance $\varepsilon \geq 0$ prior to firm realization.
- **Firm Action:** Chooses presentation $z \in \mathcal{X}$ with $\|z - X\|_2 \leq \varepsilon$ and share price $P \geq 0$.

Market Demand & Investor Heterogeneity:

- k heterogeneous investor subpopulations $j \in \{1, \dots, k\}$ with sizes N_j and valuations $V_j(z)$; total pool $\mathcal{N} = \sum_{j=1}^k N_j$.
- Traded volume under issuance cap $\eta \in (0, 1]$: $Q(z, P) = \min \left\{ \eta \mathcal{N}, \sum_{j=1}^k N_j \mathbb{1}_{\{V_j(z) \geq P\}} \right\}$.

Objectives:

- **Firm:** Maximizes total capital proceeds $\Pi(z, P) = P \cdot Q(z, P)$.
- **Regulator:** Maximizes expected social welfare $\mathcal{W}(\varepsilon) = \mathbb{E} \left[\frac{Q^*(X, \varepsilon) \cdot Y(X)}{\mathcal{N}} \right]$.

Theorem 5.2 (Convex Decomposition of Firm Optimal Response)

For fixed presentation z , optimal price satisfies $P \in \{0\} \cup \{V_j(z) : V_j(z) \geq 0\}$. The firm's joint optimization reduces to solving at most $2^k - 1$ convex programs:

$$\max_{z \in \mathcal{X}, P \geq 0} P \cdot \mathbb{E} \left[\min \left\{ \eta \mathcal{N}, \sum_{j \in S} N_j \right\} \mid X = x \right] \quad \text{s.t.} \quad \|z - x\|_2 \leq \varepsilon, P \leq V_j(z) \forall j \in S.$$

For concave valuations V_j , each subproblem is a convex program. The firm chooses the revenue-maximizing subset $S^* \subseteq \{1, \dots, k\}$.

Proposition 5.3 (Saturation beyond $\varepsilon_{\max}$)

For $\varepsilon \geq \text{diam}(\mathcal{X})$, the ball $\mathcal{B}(x, \varepsilon)$ covers all of $\mathcal{X}$. Welfare $\mathcal{W}(\varepsilon)$ saturates to a constant $\mathcal{W}(\varepsilon_{\max})$.

Interior Optimum Criterion:

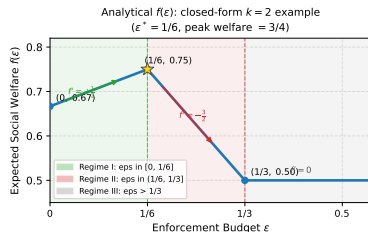
If $\mathcal{W}'(0) > 0$ and $\mathcal{W}'(\varepsilon_{\max}) < 0$, the optimal policy ε^* is strictly interior: $\varepsilon^* \in (0, \varepsilon_{\max})$ satisfying $\mathcal{W}'(\varepsilon^*) = 0$.

Decision Problem: Affine Valuations & Targeting

Bimodal Market ($k = 2$): Groups α, β with equal mass $N_\alpha = N_\beta = \mathcal{N}/2$, full supply $\eta = 1$.
Valuations: $V_\alpha(z) = w_\alpha^\top z + b_\alpha$ and $V_\beta(z) = w_\beta^\top z + b_\beta$.

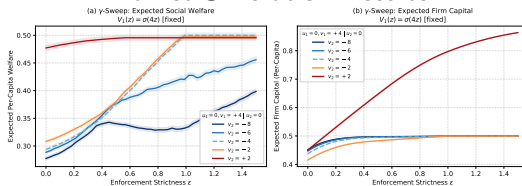
- **Target α or β alone:** $P_j^*(x, \varepsilon) = V_j(x) + \varepsilon \|w_j\|_2$, Traded Volume $Q = \mathcal{N}/2$.
- **Target both (Pooling):** $P_{\alpha\beta}^*(x, \varepsilon) = \max_{\|\Delta\|_2 \leq \varepsilon} \min\{V_\alpha(x + \Delta), V_\beta(x + \Delta)\}$, Volume $Q = \mathcal{N}$.

Analytical Welfare Curve



Analytical welfare $\mathcal{W}(\varepsilon)$ exhibiting non-monotonicity
and interior optimal tolerance ε^* .

Market Simulation Results



Welfare $\mathcal{W}(\varepsilon)$ and traded volume $Q(\varepsilon)$ vs tolerance ε ,
showing initial gains before degradation.

Unknown Market Distribution: Regulator selects tolerance $\varepsilon_t \in [0, \varepsilon_{\max}]$ over rounds $t = 1, \dots, T$; observes noisy welfare $W_t = \mathcal{W}(\varepsilon_t) + \xi_t$.

Algorithm: Matérn-5/2 GP-TS

- Maintain GP posterior over $[0, \varepsilon_{\max}]$.
- Draw inflated sample and maximize over net.
- Update posterior with realized market welfare W_t .

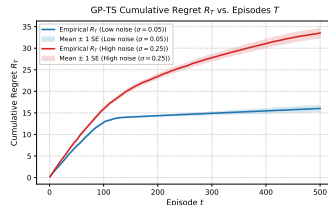
Theorem 5.5 (Online Regret Bound)

Assuming welfare $\mathcal{W} \in \mathcal{H}_k(B)$ and sub-Gaussian noise:

$$R_T = \sum_{t=1}^T [\mathcal{W}(\varepsilon^*) - \mathcal{W}(\varepsilon_t)] = \tilde{O}\left(\mathbf{T}^{2/3}\right).$$

Takeaway: Sublinear regret guarantees convergence to optimal regulatory policy despite window-dressing.

Online Learning Regret



Cumulative regret of GP Thompson Sampling compared to naive fixed tolerances, showing sublinear convergence.

Conclusion

- Chapter 2 Penalty labels and randomization alter strategic response. Learning depends on the complexity of the induced classification rules.
- Chapter 3 Imputation and regularization balance the value of side information against missing joint observations. Some information remains unidentified.
- Chapter 4 Explicit conditions identify when voluntary disclosure can improve regression. Offline estimation and online regret require their stated assumptions.
- Chapter 5 The firm problem reduces to buyer-subset programs. Regulatory learning has a conditional Matérn GP-TS regret guarantee.

Strategic response changes the decision problem.

- A penalty or a randomized rule changes the incentive to manipulate.
- A disclosure choice carries information about a sender's preferences.
- A firm's response connects regulatory tolerance to pricing and participation.

The observation structure changes what can be learned.

- Partitioned samples can leave dependencies unidentified.
- Bandit feedback reveals only the outcome of the chosen action.
- Guarantees depend on the function class, the feedback model and the comparator.

Journal Articles:

- ① **M. K. Singh** and A. A. Kulkarni. "Strategic Classification for Non-uniform Preferences using Penalty Labels and Randomisation," *Autonomous Agents and Multi-Agent Systems (JAAMAS)*, 2025.
- ② **M. K. Singh**, S. S. Kowshik, and A. A. Kulkarni. "Learning with Side Information from Partitioned Data," *Information and Inference: A Journal of the IMA* (Oxford University Press), (*Submitted*).
- ③ **M. K. Singh**, A. A. Kulkarni, and K. Mahajan. "Learning under Strategic Feature Revelation," (*Submitted to Journal*).
- ④ **M. K. Singh** and A. A. Kulkarni. "Strategic IPO Regulation under Complete and Incomplete Information," (*Submitted to Journal*).

Conference Proceedings:

- ① **M. K. Singh**, A. A. Kulkarni, and K. Mahajan. "Learning under Strategic Feature Revelation," *IEEE Conference on Decision and Control (CDC)*, 2026 (*Accepted*).
- ② **M. K. Singh** and A. A. Kulkarni. "Optimal Stochastic Decision Rule for Strategic Classification," *National Conference on Communications (NCC)*, 2024.
- ③ **M. K. Singh** and A. A. Kulkarni. "Strategic Multiclass Classification with Non-uniform Preferences and its Relation to Incentive Compatibility," *Indian Control Conference (ICC)*, 2022 (*Best Student Paper Award*).

- **Richer Strategic Models:**

- Multi-round repeated interactions and dynamic feature gaming.
- Multi-agent competition with heterogeneous decision makers.
- Non-linear hypothesis spaces with partial feature revelation.

- **Bridging Strategic Decisions & Partitioned Observations:**

- Strategic agents operating under partitioned or incomplete information structures.
- Incentive-compatible imputation mechanism design.

- **Policy & Applied Systems:**

- Regulatory design in financial credit scoring, college admissions, and hiring algorithms.
- Structurally heterogeneous and federated multi-institution learning.

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Thank You

Questions & Discussion

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The excess risk decomposes into four terms:

$$\begin{aligned} R(\hat{w}_\lambda) - R(w^*) &= \underbrace{R(\hat{w}_\lambda) - R^{\hat{\phi}}(\hat{w}_\lambda)}_{\text{(A): Imputation gap at } \hat{w}_\lambda} + \underbrace{R^{\hat{\phi}}(\hat{w}_\lambda) - R^{\hat{\phi}}(w_{\hat{\phi},\lambda}^*)}_{\text{(B): Empirical estimation}} \\ &\quad + \underbrace{R^{\hat{\phi}}(w_{\hat{\phi},\lambda}^*) - R(w_{\hat{\phi},\lambda}^*)}_{\text{(C): Imputation gap at } w_{\hat{\phi},\lambda}^*} + \underbrace{R(w_{\hat{\phi},\lambda}^*) - R(w^*)}_{\text{(D): Regularization penalty}} \end{aligned}$$

Bounding each component:

- **(A) + (C):** Imputation estimation error $\leq \min\{1/(2\lambda_z), W\} \cdot \frac{4}{\ln 2} (C_2/\sqrt{m} + L(u^*))$.
- **(B)+(D):** Bound jointly using empirical optimality and a population comparator.
- The joint contribution is $C_1/\sqrt{n} + \min\{\lambda_z |w_z^*|^2, T(v^*) - R(w^*)\}$.

The min structures in (A)+(C) and (B)+(D) arise because $\lambda_z = \infty$ automatically recovers the feature-only benchmark (no side info).

Gaussian Process with Matérn-5/2 kernel:

$$k_{5/2}(a, a') = \sigma_0^2 \left(1 + \frac{\sqrt{5}\|a - a'\|}{l} + \frac{5\|a - a'\|^2}{3l^2} \right) \exp\left(-\frac{\sqrt{5}\|a - a'\|}{l}\right)$$

Posterior updates after history $\{(a_\tau, Z_\tau)\}_{\tau=1}^{t-1}$:

$$\begin{aligned}\mu_{t-1}(a) &= \mathbf{k}_{t-1}(a)^\top (K_{t-1} + \lambda I)^{-1} \mathbf{Z}_{t-1}, \\ k_{t-1}(a, a') &= k(a, a') - \mathbf{k}_{t-1}(a)^\top (K_{t-1} + \lambda I)^{-1} \mathbf{k}_{t-1}(a').\end{aligned}$$

where $K_{t-1} = [k(a_i, a_j)]_{i,j=1}^{t-1}$ and $\mathbf{k}_{t-1}(a) = [k(a_1, a), \dots, k(a_{t-1}, a)]^\top$.

Thompson Sampling: Sample $\tilde{f}_t \sim \text{GP}(\mu_{t-1}, v_t^2 k_{t-1})$ with exploration parameter v_t , then choose action $a_t = \arg \min_a \tilde{f}_t(a)$.

If the presentation ball lies within $\mathcal{X}$, let $A = V_\alpha(x)$ and $B = V_\beta(x)$. Then the common price satisfies

$$P_{\alpha\beta}^*(x, \varepsilon) = \min_{\lambda \in [0,1]} \{ \lambda A + (1 - \lambda) B + \varepsilon \| \lambda w_\alpha + (1 - \lambda) w_\beta \|_2 \}.$$

- The minimizing λ depends on both baseline valuations and directions.
- $\lambda = 1/2$ is not implied by equal norms alone.
- If the weighted vector vanishes, the normalized-direction formula is undefined.
- When the ball meets the feature boundary, retain $x + \Delta \in \mathcal{X}$ in the primal problem.